

tiny Intrusion Detection System (tIDS)

A Behavioral Machine Learning Approach

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Outline

- 1 Introduction
- 2 Threat Model & Data Engineering
- 3 Proposed Solution: ML Architecture
- 4 Evaluation & Results
- 5 System Implementation & Demo

The Problem with Traditional IDS

- **Signature-Based Limitations:** Traditional Intrusion Detection Systems rely on pre-existing databases of known attack signatures.
- **The Blind Spot:** They operate reactively, making them inherently vulnerable to zero-day exploits and custom-crafted attacks.
- **Our Proposed Solution:** The tiny Intrusion Detection System (tIDS) shifts from signature matching to *behavioral anomaly detection* using machine learning.

We evaluated the tIDS against 7 complex network attack vectors:

- (1) STP Spoofing
- (2) ARP Poisoning
- (3) DNS Spoofing
- (4) TCP SYN Flood
- (5) Smurf Attack
- (6) IGMP Spoofing
- (7) UDP Flooding

- **Environment:** Isolated VirtualBox network — prevents attack leakage, ensures clean captures.
- **Live-Fire Attacks:** Executed 7 real-world exploits targeting Link, Network, and Transport layers.
- **Output:** 8 raw PCAP/PCAPNG files — one per attack class + Normal baseline traffic.

PCAP → CSV: Feature Extraction Pipeline

- **Tool:** `Convert_pcap_to_csv.py` (Scapy-based).
- **Method:** Traffic partitioned into **1-second time windows** — each window becomes one CSV row.
- **38 Features** extracted per window across 8 groups:
 - Traffic Volume, TCP Flags, UDP, ICMP, ARP, DNS, Control Protocols (STP/IGMP), Network Diversity.
- **Dataset:** `tIDS_final_data.csv` — 9,472 balanced instances, 8 labeled classes.

3 Integrated Modules:

- 1 `sniffer.py` — Live packet capture via Scapy; maintains rolling 1-second window.
- 2 `engine.py` — Aligns 38 features to the model's 35, scales, and classifies (<5 ms).
- 3 `main.py` — FastAPI server broadcasts predictions over WebSocket in real time.

End-to-End Flow

Capture → Extract → Scale → Predict → Broadcast

- **Latency:** Sub-second detection per cycle.
- **Control:** `/api/start` & `/api/stop` REST endpoints.

- **Preventing Overfitting:** Removed debugging columns (Timestamp, Source IP, Destination IP) so the model learns statistical behavior, not specific network nodes.
- **Variance Audit:** Extracted 38 initial mathematical features (e.g., *syn_to_ack_ratio*). Dropped 3 zero-variance features to eliminate mathematical noise.
- **Final Dataset:** 35 optimized features processed via a *StandardScaler* to normalize magnitudes.
- **Validation Strategy:** 70/30 train-test split utilizing 5-fold Cross-Validation for hyperparameter tuning.

Model Performance & Results

Classifier	Overall Accuracy	Precision (Normal Traffic)
Gaussian Naive Bayes	84%	44%
K-Nearest Neighbors (KNN)	98%	98%
Random Forest (Champion)	99%	99%

- **Naive Bayes:** Struggled with feature correlation, yielding high false positives on benign traffic.
- **KNN:** Strong geometric clustering identified stealth attacks like ARP spoofing.
- **Random Forest:** Handled the 35-dimensional space perfectly, achieving a near-zero false-positive rate (critical to prevent analyst alert fatigue).

- **Dashboard Architecture:**

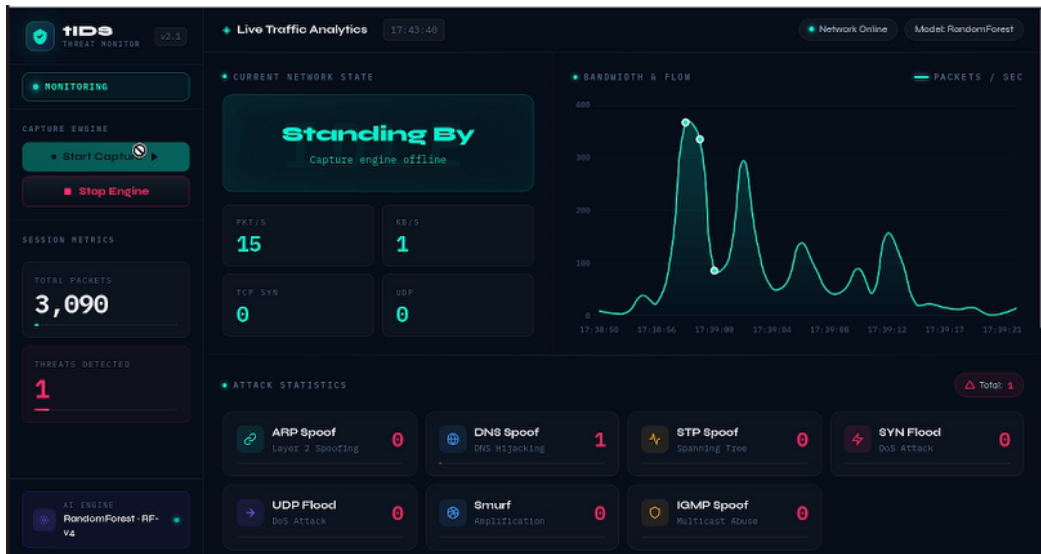
- Real-time traffic monitor with color-coded threat severity
- Live classification results showing predicted attack type + confidence score
- Timestamped alert system for detected anomalies
- Packet inspector for forensic analysis

- **ML Pipeline Integration:**

- Trained Random Forest model exported as .pkl file
- StandardScaler loaded in-memory for real-time feature normalization
- Sub-5ms inference latency per prediction window
- Queue-based architecture: Sniffer → Feature Extraction → Model → Dashboard

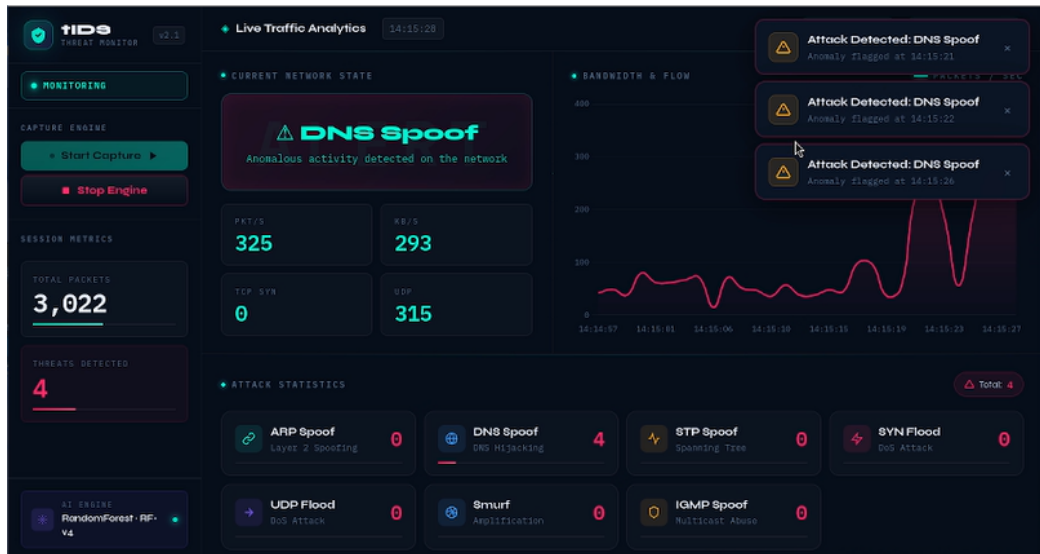
- **Detection Pipeline:** Packet Capture → Extract 35 Features → Scale Inputs → Predict (Normal/Attack) → Alert if confidence $\geq$ 70%

System Architecture



Real-time

Dashboard Walkthrough



Real-time